Life Sciences und Facility Management
ICLS Institut für Computational Life Sciences

Assignment

Assignment 2: Design of Experiments and Evaluation of a Given System

In this assignment, you will work in class with the instructors to create a simple simulation that allows for the exploration of different scenarios and parameter variations. Unlike Assignment 1, which focuses on process development and modeling, this task is centered exclusively on applying the provided simulation environment.

The goal is to derive meaningful insights into the simulated process through systematically planned experiments and a structured experiment planning matrix. The simulation data will be analyzed based on predefined KPIs and stochastic elements, whose effects you will interpret and present in both a report and a video presentation.

1. Planning of Experiments

- Develop an **experiment planning matrix** to analyze potential variations in the process.
- Determine which scenarios or parameter changes will be tested.

2. Implementation of the Experiments

- Utilize the provided simulation tool to conduct the experiments as defined in the experiment planning matrix.

3. Evaluation of Simulation Data

- Analyze the results of your simulations and present them using appropriate visualization methods (e.g., diagrams and tables).
- Interpret the obtained results and draw well-founded conclusions.

4. Documentation

- Prepare a **4-6 page report** that summarizes your approach and findings in a structured manner.
- The report should include:
  - The experiment planning matrix
  - A detailed description of the implementation
  - Achieved results
  - A well-founded interpretation of the findings